



Cloud Native Observability

Prometheus Use Case

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This article explores Prometheus use cases, showcasing its applications in infrastructure monitoring, including metrics aggregation, proactive alerting, and performance analysis.

In this lesson, we explore several Prometheus use cases, demonstrating both its origins and its practical applications in modern infrastructure monitoring.

Unified Metrics Aggregation

Organizations with multiple data centers and cloud services, such as AWS, face challenges in monitoring distributed environments. Imagine an enterprise with data centers spread across the West, Central, and East regions, alongside cloud-based services. The objective is to aggregate metrics from these diverse sources and present them in a single, easy-to-read dashboard.

Prometheus is designed to scrape metrics from various locations, whether on-premises or cloud-based, and provides built-in dashboarding utilities to display the data in a consolidated view.

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Proactive Alerting for Server Health

A common scenario for Prometheus involves monitoring server resources. Consider a MySQL database server that experiences high memory usage, potentially leading to outages. Operations teams need to be alerted when memory consumption reaches a critical level (for example, 80% usage) so they can intervene before users experience disruptions.

Prometheus offers built-in alerting mechanisms. When a metric exceeds a predefined threshold, it can trigger notifications through various channels such as email, Slack, or SMS.



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Several outages have occurred due to high memory on the server hosting a MySQL database.

Operations team would like to be notified through email when the memory gets to 80% max capacity so that proactive measure can be taken



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To configure alerting:

Define metric queries in Prometheus.

Set threshold rules.

Integrate with your preferred notification systems.

Performance Analysis for New Features

When introducing new functionalities, such as a video upload feature on a website, performance monitoring becomes crucial. For instance, if large file uploads lead to increased latency, it is important to determine the file size at which performance degrades significantly.

By collecting metrics on both average file size and request latency, Prometheus enables teams to correlate these values and pinpoint the threshold at which performance issues occur. Its visualization tools graphically represent this analysis, supporting further performance optimization.

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A new video upload feature has been added to the website, however there is some concerns about users uploading too large of videos.

The team would like to find out at which video length the applications starts to degrade and to do this, they need a chart plotting the avg file size of upload and the average latency per request

Application



Avg file size
Avg latency



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Summary of Prometheus Capabilities

Prometheus is a versatile tool that provides critical monitoring solutions, including:

Aggregated Metrics Display: Collect metrics from distributed data centers and cloud services into a unified dashboard.

Proactive Alerting: Monitor key performance indicators and trigger alerts to prevent potential outages.

Feature Performance Analysis: Analyze the impact of new application features to maintain optimal performance.

Leveraging Prometheus allows teams to maintain reliable, high-performing infrastructure by proactively identifying and addressing potential issues.

These use cases highlight how Prometheus can help organizations continuously monitor and improve their infrastructure and application performance, ensuring a robust and responsive digital environment.



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Prometheus Documentation

Monitoring Best Practices

By following this guide, you can effectively utilize Prometheus to enhance your system monitoring strategy and improve overall performance management.



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